

# Qatar-2b

R-1251

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_Qatar-2\_160323 · created 2026-07-31T11:53:24+00:00

## BENCHMARK: NON-DETECTION

### Results

|                                                   |                                   |
|---------------------------------------------------|-----------------------------------|
| Mid-Transit Time (Tmid)                           | 2457469.48767 +/- 0.00025 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.01 +/- 0.066                    |
| Transit depth (Rp/Rs)^2                           | 0.01 +/- 0.13 [%]                 |
| Orbital Inclination (inc)                         | 88.32 +/- 1.7                     |
| Airmass coefficient 1 (a1)                        | 0.72 +/- 0.33                     |
| Airmass coefficient 2 (a2)                        | 0.14 +/- 0.56                     |
| Scatter in the residuals of the lightcurve fit is | 42.28 %                           |
| Best Comparison Star                              | None                              |
| Optimal Aperture                                  | 2.23                              |
| Optimal Annulus                                   | 8.36                              |
| Transit Duration (day)                            | 0.0155 +/- 0.002                  |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R■ fit vs published    | 0.01 ± 0.066 vs 0.16327 (deviation 1.96σ) |
| Duration fit vs published | 0.0155 d vs 0.0767 d (deviation 25.77σ)   |
| Mid-time O–C              | -1925.9 min                               |
| Depth SNR                 | 0.1                                       |
| Residual scatter          | 42.28 %                                   |

### Light curve

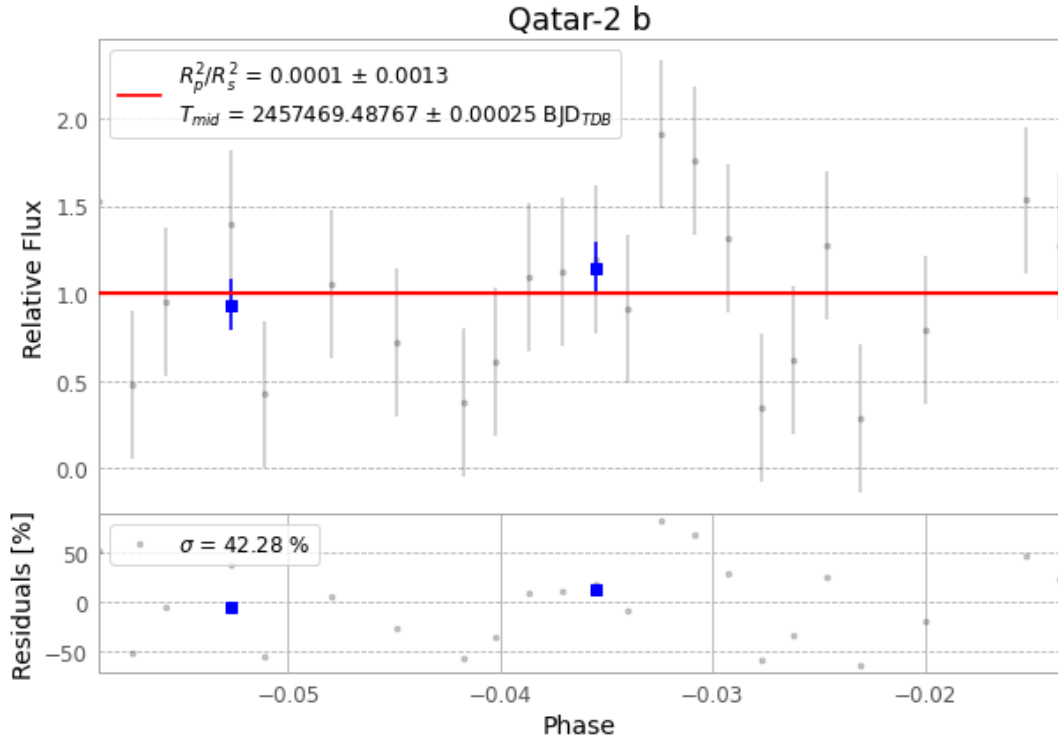


Figure 1 — FinalLightCurve\_Qatar-2 b\_2016-03-22.png (SHA-256 5d46ae12afd7b706...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [247, 470], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 7f8ee77c7136d6b5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ba3ef8b

## Data provenance

102 FITS frames (67 MB), Qatar-2160323054212.FITS → Qatar-2160323104512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-160323.log (d19f0e693e84...), FinalLightCurve\_Qatar-2 b\_2016-03-22.png (5d46ae12afd7...), FinalLightCurve\_Qatar-2 b\_2016-03-22.csv (e8091f03f5ca...)

## Compute resources

Wall time 150.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 11:53Z.